

RABBIT HOLES

Accelerate · Manipal University Jaipur

12

PHASE

GitHub as a platform

Everything that is not git: issues, projects, pages, and search done properly.



Contents

1	Where git stops	3
2	Issues	3
2.1	Linking issues to pull requests	3
2.2	Templates	4
2.3	Labels and milestones	4
3	Projects	5
4	Discussions	5
5	Pages	5
6	Gists	6
7	Search, properly	6
7.1	Finding code	6
7.2	Finding issues and pull requests	6
7.3	Finding repositories	6
8	Notifications	7
9	Things that are quietly useful	7
10	Reference	8

1 Where git stops

Git is a program for recording versions of a folder. It has no concept of a pull request, an issue, a review, a label, a milestone, or a discussion. None of those are in git and none of them ever will be.

Everything in this phase is GitHub's, built on top. That is why the workshop needed two tools: `git` to move commits, `gh` to touch anything in this document.

WHY THIS EXISTS

Keeping the boundary clear pays off constantly. When something behaves oddly, the first question is which system owns it. A rejected push is git. A pull request that will not merge is GitHub. A conflict is git. A required review is GitHub.

It also tells you what survives if you leave. Your commits are yours and clone in full. Your issues, reviews and discussions are GitHub's, and moving them means an export tool.

2 Issues

An issue is a thread with a state, a title, labels, an assignee, and a number.

```
gh issue create --title "Parser fails on blank lines" --body "..."  
gh issue list --state open --label bug  
gh issue list --assignee "@me"  
gh issue view 42  
gh issue view 42 --comments  
gh issue comment 42 --body "Reproduced on Windows too"  
gh issue close 42 --reason "not planned"  
gh issue reopen 42  
gh issue edit 42 --add-label bug --add-assignee shashwat-deep  
gh issue transfer 42 accelerate-muj/other-repo
```

2.1 Linking issues to pull requests

Put a closing keyword in the pull request description:

```
Closes #42  
Fixes #17, closes #18  
Resolves accelerate-muj/other-repo#9
```

When the pull request merges, GitHub closes those issues automatically and records the link both ways. Six months later the issue shows exactly which change resolved it, which is the kind of thing you only appreciate when you need it.

NOTE

The keyword must be in the pull request body or a commit message, not in a com-

ment added afterwards. Closes #42 typed as a comment does nothing except look like it worked.

2.2 Templates

Put templates in `.github/ISSUE_TEMPLATE/`. A YAML form gives you structured fields rather than a blank box people ignore:

```
name: Bug report
description: Something does not work
labels: ["bug"]
body:
  - type: input
    id: version
    attributes:
      label: Git version
      description: Output of `git --version`
    validations:
      required: true
  - type: textarea
    id: steps
    attributes:
      label: Steps to reproduce
      value: |
        1.
        2.
    validations:
      required: true
```

Add `.github/ISSUE_TEMPLATE/config.yml` to point people elsewhere for things that are not bugs:

```
blank_issues_enabled: false
contact_links:
  - name: Question about the workshop
    url: https://github.com/accelerate-muj/git-started/discussions
    about: Ask here instead of opening an issue
```

2.3 Labels and milestones

```
gh label create "good first issue" --color 7057ff --description "Good for newcomers"
gh label list
gh issue list --label "good first issue"
```

good first issue and help wanted are special: GitHub surfaces repositories using them in its own discovery pages. If you want outside contributors, they are worth using accurately rather than decoratively.

Milestones group issues toward a target, and show a progress bar. Useful for a release or an event, less useful as a permanent bucket.

3 Projects

Projects are spreadsheets and boards over issues and pull requests, with custom fields you define: status, priority, effort, iteration, or anything else.

```
gh project list --owner accelerate-muj
gh project item-list 3 --owner accelerate-muj
gh project item-add 3 --owner accelerate-muj --url <issue-url>
```

The useful part is built-in automation: an item moves to Done when its pull request merges, or to In Progress when someone is assigned. Set it up once and the board stops needing manual maintenance, which is the only reason boards survive.

NOTE

Projects are per user or per organisation, not per repository, so one board can span several repositories. That is the main reason to use them over a simple label scheme.

4 Discussions

Issues are for things with a resolution. Discussions are for things without one: questions, ideas, showing off what you built.

Categories can be Q&A, where answers can be marked accepted, which makes Discussions a decent support forum for a club or a small project.

```
gh api repos/:owner/:repo/discussions --jq '.[].title'
```

Enable them under repository settings. For a workshop repository they are worth having, because otherwise every question becomes an issue and the issue list stops being a list of work.

5 Pages

Free static hosting from a repository.

```
gh api -X POST repos/:owner/:repo/pages -f source[branch]=main -f source[path]=/docs
gh browse --settings
```

Three ways to publish: a branch, a /docs folder, or a GitHub Actions workflow. The Actions route is what any static site generator uses, and it is covered in Phase 14.

Add a CNAME file containing your domain to use a custom one. HTTPS is automatic.

6 Gists

A gist is a tiny repository for a snippet. It has full git history and can be cloned, which people forget.

```
gh gist create script.sh --public --desc "Bisect helper"
gh gist create --filename notes.md -
gh gist list
gh gist edit <id>
gh gist clone <id>
```

TRAP

A secret gist is unlisted, not private. Anyone with the URL can read it, and the URL is not protected. Never put credentials in one.

7 Search, properly

GitHub's search syntax is genuinely powerful and almost nobody learns it.

7.1 Finding code

```
language:python "def parse_sonnet"
repo:accelerate-muj/git-started path:*.tex rerere
org:accelerate-muj filename:.gitattributes
"force-with-lease" NOT test language:markdown
/git\s+rebase\s+--onto/
```

The last one is a regular expression, which the current code search supports. Also useful: `symbol:functionName` finds definitions rather than every mention.

7.2 Finding issues and pull requests

```
is:issue is:open label:bug sort:created-desc
is:pr is:merged author:@me
is:pr review-requested:@me is:open
is:pr is:open draft:false -author:@me
is:issue no:assignee label:"good first issue"
is:pr merged:>2026-08-01
is:issue involves:@me updated:>2026-08-20
```

`review-requested:@me is:open` is the one to bookmark. It is your actual queue, and it is not the same as your notifications.

7.3 Finding repositories

```
stars:>1000 language:rust pushed:>2026-01-01
org:accelerate-muj archived:false
topic:workshop topic:git
```

From the terminal:

```
gh search code "force-with-lease" --owner accelerate-muj
gh search issues --label bug --state open --owner accelerate-muj
gh search prs --review-requested @me --state open
gh search repos --topic workshop --sort stars
```

8 Notifications

The default is to notify on everything you are subscribed to, which quickly becomes noise that gets ignored entirely.

```
gh api notifications --jq '.[ ] | "\(.subject.type) \(.repository.full_name) \(.subject.title)'"
gh api -X PUT notifications -f read=true
```

Worth doing in the web interface once: set watching to **Participating and mentions** for repositories you contribute to occasionally, and **All activity** only for ones you maintain. @mentions and review requests still reach you either way.

9 Things that are quietly useful

Keyboard shortcuts Press `?` on any page. `t` opens the file finder in a repository, `l` jumps to a line, `s` focuses search, and `.` opens the repository in a web editor.

Press the full stop key Genuinely. `.` in any repository opens VS Code in the browser, editing that repository. `github.dev` in the URL does the same.

Blame from the browser `b` on a file view. Combined with `.git-blame-ignore-revs` from Phase 6 it is better than most local tools.

Permalinks Press `y` on a file view and the URL rewrites to a specific commit. Without it, a line link rots the moment the file changes.

Suggested changes In a review, use a `suggestion` code block and the author can apply your edit with one click.

Draft pull requests `gh pr create --draft` marks work as in progress. Reviewers are not requested until you mark it ready.

CODEOWNERS Automatically requests review from the right people per path. Phase 15.

Dependabot Opens pull requests for outdated or vulnerable dependencies. Add `.github/dependabot.yml`.

Insights The repository's Insights tab has contributor graphs, traffic, and a dependency graph. Traffic is the only place to see clone and view counts.

10 Reference

Goal	Command or syntax
Close an issue when a PR merges	<code>Closes #42</code> in the PR body
My review queue	<code>is:pr review-requested:@me is:open</code>
Unassigned beginner issues	<code>is:issue no:assignee label:"good first issue"</code>
Search code in an org	<code>gh search code "text" --owner <org></code>
Open the web editor	press <code>.</code> in any repository
Permalink to a line	press <code>y</code> on the file view
Create a draft PR	<code>gh pr create --draft</code>
Structured issue forms	<code>.github/ISSUE_TEMPLATE/*.yml</code>
Redirect questions elsewhere	<code>ISSUE_TEMPLATE/config.yml</code>

CHECK YOURSELF

1. Which things in this phase exist in git itself? (Careful.)
2. Why does `Closes #42` in a comment not close the issue?
3. What is the difference between a secret gist and a private one?
4. Write the search that finds open pull requests waiting on your review.
5. When should something be a Discussion rather than an Issue?

Next: Phase 13, *The gh CLI in full*, including `gh api`, which reaches every part of GitHub that has no dedicated command.